BASHVS: A Multispectral and SAR Image Fusion Method Based on Bidirectional Aggregation of Saliency in Human Visual System

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***Abstract*— The limitations of remote sensing sensor technology make it difficult to simultaneously capture earth observation information presented in different forms within a single remote sensing image. Acquiring more plentiful target information through fusion technology has therefore remained a research hotspot. The fusion of multispectral (MS) and synthetic aperture radar (SAR) imagery integrates spectral and backscatter information, thereby improving land cover (LC) classification effects. However, current pixel-level fusion methods often fail to adequately account for the model differences between SAR and MS images, leading to spectral–spatial inconsistencies and severe degradation from speckle noise. To address this problem, a fusion method is proposed based on bidirectional aggregation of saliency in the human visual system (BASHVS). First, the SAR and MS images are decomposed into base and detail layers using a synchronized-anisotropic diffusion algorithm. Subsequently, the detail layer is fused using a new sum of modified anisotropic Laplacian (NSMAL) algorithm. Finally, for base layer fusion, pixel saliency (PS) and structural saliency are extracted bidirectionally. The BASHVS is compared with 16 existing fusion methods using ten evaluation metrics. The results demonstrate that BASHVS achieves the best comprehensive performance and significantly improves the visual quality of the fused images. LC classification using BASHVS fused images shows an average increase of 1.050% in overall accuracy (OA) and 0.014 in Kappa coefficient compared to the original MS images, confirming its advantage for LC classification. The source code of BASHVS is shared at https://github.com/CHUANGL8346/BASHVS.**

***Index Terms*— Bidirectional aggregation, human visual system (HVS), multiscale perceived contrasts, multispectral (MS) and synthetic aperture radar (SAR) image fusion, visual saliency (VS).**

I. Introduction

With the development of diverse earth observation technologies, a single image can no longer meet the technical requirements. Therefore, fusion methods for remote sensing imagery tailored to different application scenarios have been proposed to augment image information [1], [2]. Multispectral (MS) and synthetic aperture radar (SAR) images are complementary and exhibit structural consistency. The fusion of these two modalities, commonly referred to as heterogeneous image fusion, is a type of multisource data fusion that aims to mitigate the heterogeneous differences between modalities while preserving the specific semantic integrity of each modality [3]. SAR emits microwave radiation capable of penetrating clouds, haze, dust, and most climatic conditions except heavy rainfall. This enables SAR to collect the backscattered energy from ground objects under all-weather conditions and at any time. This backscattered energy reflects the structural features of various ground objects [4]. Hence, SAR imagery provides rich spatial information and captures distinct structural features of ground objects. However, SAR imagery is affected by speckle noise and lacks spectral information, which compromises its interpretability. MS sensors are sensitive to specific wavelength bands, enabling them to detect electromagnetic radiation across different wavelengths. Consequently, MS imagery offers superior spectral resolution [5]. However, the spectral information in MS images is derived from sunlight reflected by ground objects. Critically, different ground objects may exhibit similar spectral signatures. This complicates remote sensing interpretation. Therefore, fusing SAR and MS images enables the information of the imaged region to be complete, and is applied to tasks such as crop mapping, land cover (LC) classification, and urban impervious surface extraction [6], [7], [8], [9], [10]. (An example illustrating this complementary nature for the same land use type is shown in Fig. 1.)

Heterogeneous image fusion is challenging due to the significant differences in imaging mechanisms and spectral characteristics between the source images [11]. Pixel-level

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(a) (b)

Fig. 1. Pair of SAR and MS images having the same land use type. (a) SAR image. (b) MS image.

fusion methods offer advantages for such tasks, as they preserve spatial details, complement information content, enhance spatial details, and reduce interpretation uncertainty, thereby improving visual quality and decision reliability. Pixel-level fusion methods can be broadly categorized into four main types: the component substitution (CS) method, the multiscale decomposition (MSD) method, the variational optimization (VO)-based method, and the hybrid method [12]. CS replaces a component (often representing spatial information) within the MS image with a high spatial-resolution image [13]. While CS effectively enhances spatial details, it often introduces significant spectral distortion [5]. MSD decomposes the source images into small-to-large scales. Corresponding sub-bands undergo level-specific fusion rule application. The final composite is synthesized via inverse multiscale transformation of integrated frequency components [1]. MSD-based fusion techniques include pyramid-based, wavelet-based, and multiscale geometric analysis-based methods [14]. MSD methods generally preserve spectral information well and enable layered information extraction. However, they may lose fine spatial details and can be susceptible to amplifying speckle noise present in the SAR image [14]. VO-based methods formulate image fusion as an image restoration problem. An interimage correlation model is established to derive an energy function, yielding the fused image through optimization [11]. These approaches primarily encompass regularization-based optimization, Bayesian inference, and sparse representation-based optimization [15]. VO-based methods can mitigate registration errors to some extent. However, their high computational complexity poses a significant drawback for applications requiring real-time processing. Hybrid methods aim to combine the strengths of VO-based, CS, and MSD approaches to achieve more robust fusion [16]. However, designing effective hybrid methods involves greater complexity. Poorly designed integration can lead to the loss of key information or introduce information redundancy. Recently, deep learning methods based on the frequency domain [17], Mamba model [18], masked image modeling (MIM) [19], and contrastive learning [20] have shown good performance in heterogeneous remote sensing image fusion tasks. However, deep learning methods exhibit limited generalization ability, requiring further improvement for broader applicability [21]. Methods that integrate deep learning with VO will be future choices for heterogeneous image fusion [22].

Although numerous fusion methods exist, recent research has focused on developing hybrid methods for fusing SAR and MS images. Notably, Schmitt et al. [23], [24] released a benchmark dataset comprising co-registered Sentinel-1 (SAR) and Sentinel-2 (MS) satellite images, specifically designed for MS-SAR fusion research. Early attempts utilizing this dataset often employed CS methods. While CS demonstrated the feasibility of SAR-MS fusion, it inevitably introduced significant spectral distortion [25]. Given MSD's superior ability to preserve spectral fidelity and its capacity to extract multiscale information beneficial for applications, the combination of CS and MSD has gained popularity for fusing MS and SAR images [26], [27]. Gong et al. [28] employed MSD within their framework to address limitations in existing methods, such as insensitivity to edge information, susceptibility to speckle noise, suboptimal fusion rules, artifacts in decomposed images, and constraints in spatial/scale feature separation. However, a critical drawback is that MSD can readily introduce the speckle noise inherent in SAR images within the fusion result. Therefore, effective noise suppression constraints and optimization strategies are crucial. Furthermore, irrational decomposition rules can lead to redundant or inefficient representation of information across scales. Consequently, VO has emerged as a promising approach and is increasingly being integrated with CS and MSD in hybrid fusion frameworks [29], [30], [31]. For example, by leveraging visual saliency (VS) features as constraints, Ye et al. [16] achieved improved visual quality and mitigated the effects of SAR speckle noise in the fused result. Nevertheless, two key issues persist with these hybrid approaches: 1) they extract and integrate features from MS and SAR images without adequately accounting for the fundamental differences in their imaging mechanisms, leading to information inconsistency and significant interference from SAR speckle noise and 2) the MSD component within these hybrid frameworks still fails to adequately resolve the problem of information redundancy across scales.

In response to the above issues, the proposed fusion method, based on synchronized-anisotropic diffusion (SAD) and bidirectional aggregation of the saliency of the human visual system (HVS), meets the following requirements: the base layer and detail layer are complementary, and salient features are extracted and integrated on a consistent basis. To verify the effectiveness of the BASHVS, first, we compare it under ten evaluation indicators with 16 fusion methods. Then, ablation experiments are designed to indicate the advantages of the decomposition rules and fusion rules of this method. Finally, we perform LC classification experiments to verify if the BASHVS can be applied in practice. The BASHVS has the following three contributions.

1) *Synchronous MSD:* The synchronous processing attribute of the SAD is used for MSD and sufficient extraction of energy information and texture information. So, these two kinds of information

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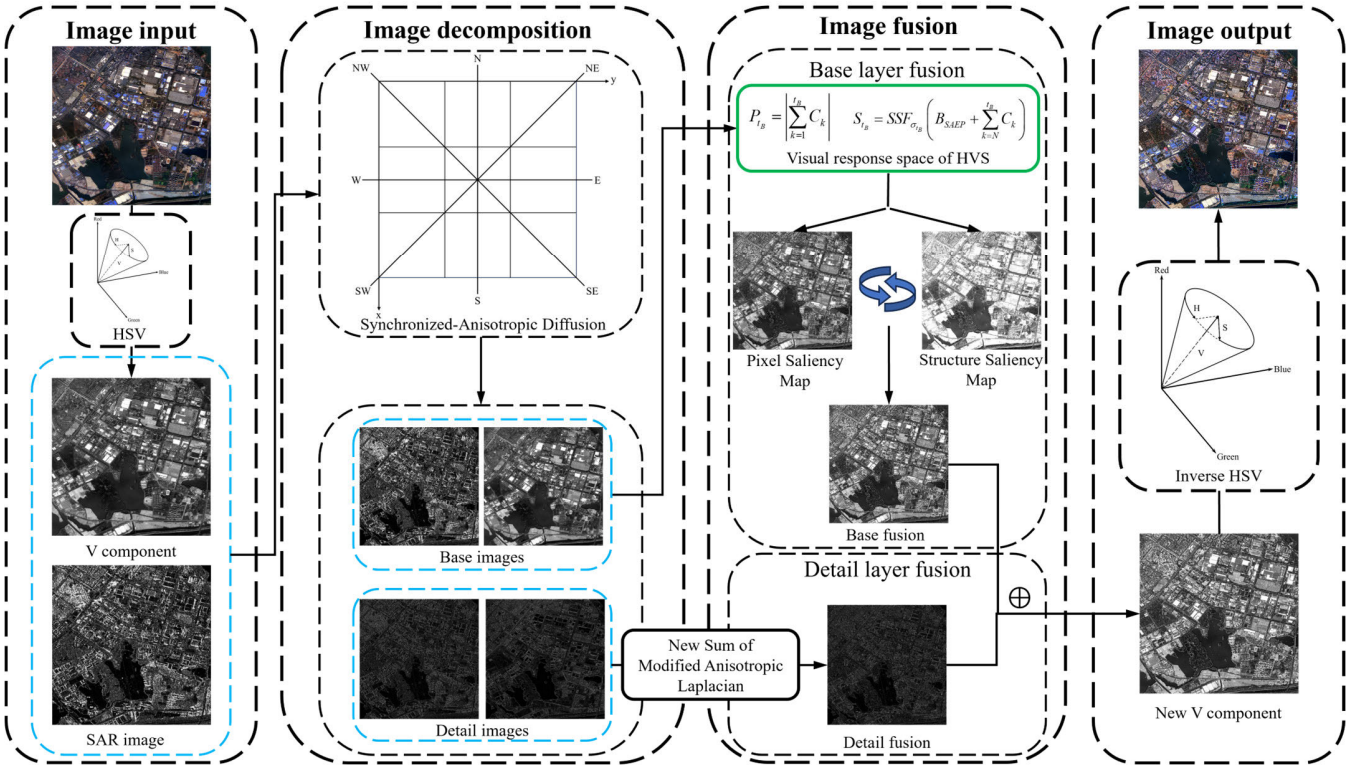


Fig. 2. Fusion framework for MS and SAR images of this article.

on MS and SAR images have a stronger relevance. In addition, this algorithm can ignore irrelevant information.

2) *Bidirectional Aggregation of Saliency in the Same Perceptual State:* A bidirectional aggregation of the saliency of the HVS algorithm, which considers model differences, is proposed to provide the same perceptual state to collect the saliency feature of the MS and SAR image, so the influence of the external physical factors (for instance, the imaging mechanism of the sensor and the brightness of the environment, etc.) can be reduced.
3) *Context Association Between Decomposition and Detail Layer Fusion:* The new sum of modified anisotropic Laplacian (NSMAL), which contains the diffusion function values generated from the decomposition process, is utilized as a fusion rule of texture information, making a context association between the decomposition process and the fusion process. This process sufficiently fuses the texture information decomposed from MS and SAR, avoiding the introduction of superfluity.

In Section II, the SAD algorithm, the bidirectional aggregation of the saliency of the HVS algorithm, and the NSMAL algorithm are introduced in detail. Section III specifically introduces the experiment contents, including datasets, parameters, evaluation indicators, qualitative evaluations, quantitative evaluations, ablation experiments, and LC classification experiments. Section IV provides a summary of the research.

II. Methodology

The operational principles of BASHVS will be delineated in this part. In the image decomposing process, an SAD algorithm is used to simultaneously extract the base layer and the detail layer from the SAR and MS images. In the image fusion section, an NSMAL algorithm is introduced, and we propose a bidirectional aggregation of the saliency of the HVS algorithm. The specific process is shown in Fig. 2.

1) *HSV Transform:* First, we use the Hue–Saturation–Value algorithm [32] to decouple the hue, saturation, and value components from the MS image. These three kinds of components are all presented in the form of images.
2) *Synchronized-Anisotropic Diffusion:* The SAD algorithm is used to vaguely calculate the value component and the SAR image, acquiring the base layer. The base layer is separately subtracted from the source image, acquiring the detail layer. Besides, the function values of eight directions are acquired in the decomposition process, which can be utilized to fuse the detail layer.
3) *Bidirectional Aggregation of the Saliency in the Same Perceptual Space:* An algorithm that bidirectional extracts the saliency of structure and pixel in the same perceptual space is used to guide the fusion process of the base layer to reduce the influence of the external physical factors and augment fidelity preservation in energy.
4) *NSMAL:* For the detail layer, a modified Anisotropic Laplacian (NSMAL) algorithm is used to fuse the detail layer. The function values of eight directions are used to make an effective and irredundant fusion of the detailed information of the value component and the SAR image.
5) *HSV Inverse Transform:* Finally, we have a linear summation of the fusion results of the base layer and the detail layer to gain the new value component and use the Hue–Saturation–Value algorithm [32] to inverse operate

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the new value component, the hue component and the saturation component, acquiring the new MS image.

A. Image Decomposition by Synchronized-Anisotropic Diffusion

The anisotropic diffusion means to uses the partial differential equation (PDE) to smooth the area of the input image with small changes in grayscale value and simultaneously retain the area with a mutation in grayscale value, i.e., the edge [33]. This theory regards an image as a temperature field and regards each pixel as the heat flux [34]. First, the SAR image is concatenated with the MS image, and this step is called "synchronism"; then, the anisotropic diffusion algorithm is used in the decomposition process of the concatenated image. This algorithm effectively preserves image edge definition and textural details while enhancing interlayer coherence between structural foundations and high-frequency components. The anisotropic diffusion algorithm is used to calculate the image obtained by the $(t-1)$th iteration and get the new image. The position (x, y) in the new image has the pixel value $I^t_{x,y}$, which is expressed as

$$\begin{aligned} I^t_{x,y} = I^{t-1}_{x,y} + \lambda \big(& c^{t-1}_{Nx,y} \cdot \overline{\nabla}_N I^{t-1}_{x,y} + c^{t-1}_{\mathrm{NE}x,y} \cdot \overline{\nabla}_{\mathrm{NE}} I^{t-1}_{x,y} \\ & + c^{t-1}_{Ex,y} \cdot \overline{\nabla}_E I^{t-1}_{x,y} + c^{t-1}_{\mathrm{SE}x,y} \cdot \overline{\nabla}_{\mathrm{SE}} I^{t-1}_{x,y} \\ & + c^{t-1}_{Sx,y} \cdot \overline{\nabla}_S I^{t-1}_{x,y} + c^{t-1}_{\mathrm{SW}x,y} \cdot \overline{\nabla}_{\mathrm{SW}} I^{t-1}_{x,y} \\ & + c^{t-1}_{Wx,y} \cdot \overline{\nabla}_W I^{t-1}_{x,y} + c^{t-1}_{\mathrm{NW}x,y} \cdot \overline{\nabla}_{\mathrm{NW}} I^{t-1}_{x,y} \big) \end{aligned} \tag{1}$$

where $\lambda \in [0, 1/4]$, and it can gain more stable calculation results of (1). To cooperate with the fusion process of the base layer, the value of λ is set to 1/30 000, and the setting reason will be discussed in Section III-A. The subscripts N, NE, and NW, etc., respectively, represent eight directions. The mark $\overline{\nabla}$ represents the nearest neighbor difference. $c^{t-1}_{Nx,y}$, $c^{t-1}_{Ex,y}$, $c^{t-1}_{Sx,y}$, $c^{t-1}_{Wx,y}$, $c^{t-1}_{\mathrm{NE}x,y}$, $c^{t-1}_{\mathrm{SE}x,y}$, $c^{t-1}_{\mathrm{SW}x,y}$, and $c^{t-1}_{\mathrm{NW}x,y}$, respectively, represent the diffusion function $c(x, y, t) = g(\|\nabla I(x, y, t)\|)$ in their corresponding directions. The function $g(\cdot)$ is a monotonically decreasing function and $g(0) = 1$. $g(\cdot)$ is an arbitrary function that can maintain a balance between the edge's smoothness and maintenance. These functions can be integrated into [34]

$$g(\nabla I, K, \alpha) = \frac{1}{1 + \left(\frac{\|\nabla I\|}{K}\right)^{\alpha+1}} \tag{2}$$

where a larger weight would be given to the larger area, and a smaller weight would be given to the smaller area in the calculation. $\alpha > 0$. K is called the edge threshold. To effectively retain the useful information of the SAR and the MS imagery, K and α are, respectively, set to 20 and 1 [35].

The whole calculation process is the SAD, and the base layer can be expressed as the following formula:

$$B^t_I = \mathrm{SAD}(I) \tag{3}$$

where B^t_I is the calculation results of the tth anisotropic diffusion, i.e., the base layer. The number of iterations t is set to 30 [35]. I represents the input image. The detail layer is derived through differential computation

$$D^t_I = I - B^t_I. \tag{4}$$

In summary, the decomposition method based on the SAD model can be summarized as Algorithm 1.

Algorithm 1 A Decomposition Process Based on Synchronized-Anisotropic Diffusion

Input: I_{SAR}, I_{MS}, t, K, α, λ.
Output: B_{SAR}, B_{MS}, D_{SAR}, D_{MS}, c_N, c_{NE}, c_E, c_{SE}, c_S, c_{SW}, c_W, c_{NW}.
1: Utilize the concatenate method to generate a "concatenate image" I.
2: **for** t = 1: iteration values **do**
3: Use Equations (1) and (2) to construct the SAD model.
4: Use Equation (3) to gain the diffusion function values of all directions and the base layer of the "concatenate image."
5: **end for**
6: Use Equation (4) to gain the detail layer of the "concatenate images."
7: Segment B^t_I to gain the base layer of the SAR and the MS image: B_{SAR}, B_{MS}.
8: Segment D^t_I to gain detail layer of the SAR and the MS image: D_{SAR}, D_{MS}.

B. Base Layer Fusion by Bidirectional Aggregation of Saliency of HVS

1) Multiple Scales Perceptual Contrast in the Same Response Space: A bidirectional aggregation of the saliency in the HVS (SHVS) algorithm, which is used to calculate the saliency of the SAR and the MS image in the same perceptual state, is proposed. First, scale-aware edge-preserving (SAEP) filtering [36] is used to, respectively, run the iterative calculations of B_{MS} and B_{SAR}, obtaining different spatial frequency (SF) bands. As HVS has a characteristic of local adaptation, the feature of contrast sensitivity of HVS can be simulated by calculating the band-limited local contrast through $R_{t_B} = D_{t_B}/(I_{t_B} + \kappa)^{\alpha}$ and performing a local adaptation of the background luminance at each position. I_{t_B} represents the result of the t_Bth SAEP calculation. The low-pass image I_{t_B} is the local average luminance in the contrast calculation. $D_{t_B} = I_{t_B-1} - I_{t_B}$ represents the bandpass layer obtained by the t_Bth SAEP calculation. The iteration time t_B is set to 4 in this article. κ is a small positive constant. When calculating the band-limited local contrast of the MS and the SAR image, α is, respectively, set to 0.4 and 0.8, increasing the contrast sensitivity. Then, a piecewise nonlinear transfer function is used to transfer each frequency band to the HVS perceptual sensitivity domain

$$V_{t_B} = \begin{cases} \omega \mathrm{sgn}\left(R_{t_B}\right)\left(R_{t_B}/\mu\right)^{0.5}, & \left|R_{t_B}\right| \geq \mu \\ \omega \mathrm{sgn}\left(R_{t_B}\right)\left(R_{t_B}/\mu\right)^{t_V}, & \left|R_{t_B}\right| < \mu \end{cases} \tag{5}$$

where $\mathrm{sgn}(\cdot)$ denotes the sign function, ω a constant, and t_V the exponent for the frequency band transfer function. To simulate the constant of the super-threshold contrast, as the transfer function converges (i.e., $|R_{t_B}| \geq \mu$), μ can control the inputted contrast. ω is set to 22 [36]. At various levels of scale, t_V is separately set to 1.40, 1.15, 1.04, and 1.15 [37], simulating the situation that, at relatively high and low spatial frequencies,

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Fig. 3. Aggregation results of perceived contrasts with different μ. (a) $\mu = 0.2$, (b) $\mu = 0.4$, (c) $\mu = 0.6$, (d) $\mu = 0.8$, (e) $\mu = 1.0$, and (f) $\mu = 1.2$.

they all have a decrease in the contrast sensitivity of HVS. The aggregation results of perceived contrasts with different μ are shown in Fig. 3. It can be seen that a low value of μ would make the grayscale value of high-reflectivity targets be enhanced, so the targets with low grayscale value are covered up. A high value of μ would suppress the perceived contrast. In comparison, the value of μ is set to 0.6. To suppress the noise of small-scale layers and the intensity saturation phenomenon of large-scale layers, in the small-scale layers ($t_B \leq 2$), a threshold τ is set for the perceived contrast, and the contrast less than the threshold is set to 0:

$$C_{t_B} = \begin{cases} 0, & \text{if } \left|V_{t_B}\right| < \tau \\ V_{t_B}, & \text{otherwise} \end{cases}, \quad t_B = 1, 2.$$

As poor illumination conditions are more likely to produce image noise, $\tau = 4(1 - \overline{I_0}/255)^2/t_B$. $\overline{I_0}$ is derived as the mean of I_0 and specifies the illumination condition. For the perceived contrast of the large-scale layers, an adaptive compression method is used to inhibit the intensity saturation phenomenon:

$$C_{t_B} = \begin{cases} \eta V_{t_B}, & \text{if } V_{t_B} > 0 \\ V_{t_B}, & \text{otherwise} \end{cases}, \quad 2 < t_B < N$$

$\eta = 0.7(I_0/255)^{1.5} + 0.3$ is the adaptive compression ratio calculated according to the source image I_0. The compression ratio's dynamic range is constrained to 0.7, with a minimum value of 0.3 [36].

2) Bidirectional Aggregation of Saliency: Within the HVS perceptual domain, pixel and structural saliency are derived via bidirectional saliency propagation. Summing the multiscale perceived contrasts of this scale-space directly to gain the pixel saliency (PS) of t_B in each scale level

$$P_{t_B} = \left|\sum_{k=1}^{t_B} C_k\right|, \quad t_B = 1, 2, \ldots, N \tag{6}$$

where the calculation of the PS aggregation begins with the first scale level and ends with the Nth scale level. This method can compute the PS of each scale through the contrast of multiple scales, which helps to retain a more stable trans-scale saliency contrast feature [36].

For the structural saliency, calculate in the direction of the base layer to a higher scale level, which is opposite to the direction of calculating PS

$$S_{t_B} = \mathrm{SSF}_{\sigma_{t_B}}\left(B_{\mathrm{SAEP}} + \sum_{k=N}^{t_B} C_k\right), \quad t_B = 1, 2, \ldots, N \tag{7}$$

where $\mathrm{SSF}_{\sigma_{t_B}}(\cdot)$ is the function to calculate the structural saliency, and σ_{t_B} is the scale parameter. Multiscale structural saliency measurement [38] is utilized to calculate $\mathrm{SSF}_{\sigma_{t_B}}(\cdot)$, and $\sigma_{t_B} = (r_{t_B,0} + r_{t_B,1})/2$ is defined based on the discrete scales. In (7), the local-to-global feature merging can help aggregate relevant features at a lower scale level, which provides a great foundation for the structural saliency calculation [36]. Then, the signs $B^{\mathrm{SAR}}_{\mathrm{SAEP}}$ and $B^{\mathrm{MS}}_{\mathrm{SAEP}}$ are used to represent the last level results of SAEP processing for the SAR image and MS image, respectively, and the fusion result is $B^{f}_{\mathrm{SAEP}} = (1 - W^{\mathrm{ms}})B^{\mathrm{SAR}}_{\mathrm{SAEP}} + W^{\mathrm{ms}}B^{\mathrm{MS}}_{\mathrm{SAEP}}$, W^{ms} represents the fusion weight of the MS image.

The PS and the structure saliency (SS) are transferred to the physical domain through the following formula:

$$\mathrm{PS}_{t_B-1} = \mathrm{RPS}_{t_B} \times \left(\mathrm{PS}_{t_B} + \kappa\right)^{\alpha} + \mathrm{PS}_{t_B} \tag{8}$$

$$\mathrm{SS}_{t_B-1} = \mathrm{RSS}_{t_B} \times \left(\mathrm{SS}_{t_B} + \kappa\right)^{\alpha} + \mathrm{SS}_{t_B} \tag{9}$$

where PS_{t_B} and SS_{t_B} represent the PS and the structural saliency obtained by each step of the inverse transformation. PS_4 and SS_4 are all set to B^{f}_{SAEP}, $t_B = 1, 2, 3, 4$. The obtained PS_0 and SS_0 are the PS and the structural saliency of the final physical domain.

$$\mathrm{RPS}_{t_B} = \begin{cases} \mu \mathrm{sgn}\left(P_{t_B}\right)\left(\left|P_{t_B}\right|/\omega\right)^2, & \text{if } P_{t_B} \geq \omega \\ \mu \mathrm{sgn}\left(P_{t_B}\right)\left(\left|P_{t_B}\right|/\omega\right)^{1/t_V}, & \text{otherwise} \end{cases}$$

and

$$\mathrm{RSS}_{t_B} = \begin{cases} \mu \mathrm{sgn}\left(S_{t_B}\right)\left(\left|S_{t_B}\right|/\omega\right)^2, & \text{if } S_{t_B} \geq \omega \\ \mu \mathrm{sgn}\left(S_{t_B}\right)\left(\left|S_{t_B}\right|/\omega\right)^{1/t_V}, & \text{otherwise} \end{cases}$$

represents recursive computations to the saliency of the corresponding t_B scale level. The recursive computation begins with $t_B = 4$ and ends with $t_B = 1$. Finally, $\mathrm{PSM} = wB_{\mathrm{MS}} + (1 - w)B_{\mathrm{SAR}}$ is used to fuse the PS of the SAR and the MS image, where PSM is the PS Map. B_{SAR} and B_{MS} are the SAD decomposed base layer image, and

$$w = \begin{cases} 1, & \text{if } \mathrm{PS}^{\mathrm{MS}}_0 > \mathrm{PS}^{\mathrm{SAR}}_0 \\ 0, & \text{otherwise} \end{cases}$$

is the fusion weight. To dynamically adjust the contribution ratio of B_{MS} and B_{SAR} in calculating the structural saliency map (SSM), a hybrid conditional weighted fusion method $\mathrm{SSM} = (2 - k_1 \cdot k_2)B_{\mathrm{MS}} + (k_1 \cdot k_2)B_{\mathrm{SAR}}$ is used to fuse the structural saliency of the SAR and the MS image, where SSM is the structural saliency map.

$$k_1 = \begin{cases} 1, & \text{if GW} = 1 \\ 0.5, & \text{otherwise} \end{cases}$$

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$$k_2 = \begin{cases} 1, & \text{if } B_{\text{MS}} < B_{\text{SAR}} \\ 0, & \text{otherwise} \end{cases}, \quad \text{and}$$

$$\text{GW} = \begin{cases} 1, & \text{if } \text{SS}_0^{\text{SAR}} > \text{SS}_0^{\text{MS}} \\ 0, & \text{otherwise.} \end{cases}$$

3) Constraints on Pixel and Structural Features: PSM and SSM are processed by an improved total variation model to ensure the fusion result exhibits pixel intensity similar to PSM and gradient similar to SSM simultaneously [16], [39]

$$y^* = \arg\min_y \left\{ \sum_{i=1}^{mn} |y_i - (\text{PSM}_i - \text{SSM}_i)| \delta J(y) \right\}, \quad \text{with}$$

$$J(y) = \sum_{i=1}^{mn} |\nabla_i y| = \sum_{i=1}^{mn} \sqrt{\left(\nabla_i^v y\right)^2 + \left(\nabla_i^h y\right)^2} \tag{10}$$

where $y = x - \text{SSM}$, x is the fused output of PSM and SSM, δ is a strictly positive coefficient that controls the balance between the two terms, and ∇ is the gradient operator that is used to calculate the gradient information of fusion results and SSM. A weighted linear combination integrating the fused base layer $B_F = x$ and detail layer D_F (i.e., $V' = B_F + D_F$) is used to obtain a new value component V' which includes the information of the MS and the SAR image simultaneously. Finally, an inverse transform of the color space of the new value component V', the hue (H), and the saturation (S) can be used to gain the new MS image.

C. Detail Layer Fusion by NSMAL

A NSMAL algorithm is used to fuse the detail layer of the SAR and the MS image. This algorithm needs to add the diffusion function values produced by the SAD algorithm in the decomposition process, associating decomposition with fusion and fusing the detailed information of these two images effectively and without redundancy. The new improved anisotropic Laplacian operator is represented as

$$\begin{aligned} &\text{NSMAL}(a,b) \\ &= |(c_N + c_S) I(a,b) - c_N I(a-T,b) - c_S I(a+T,b)| \\ &\quad + |(c_W + c_E) I(a,b) - c_W I(a,b-T) - c_E I(a,b+T)| \\ &\quad + |(c_{\text{NW}} + c_{\text{SE}}) I(a,b) - c_{\text{NW}} I(a-T,b-T) \\ &\qquad - c_{\text{SE}} I(a+T,b+T)| \\ &\quad + |(c_{\text{NE}} + c_{\text{SW}}) I(a,b) \\ &\qquad - c_{\text{NE}} I(a-T,b+T) - c_{\text{SW}} I(a+T,b-T)| \end{aligned} \tag{11}$$

where c_N, c_{NE}, and c_{NW}, etc., respectively, represent the diffusion function values. (a, b) is the position of each pixel. T represents the variable spacing between pixels, which improves the robustness of operators when the grayscale value has a large change span. The final enhanced algorithm takes the form

$$\text{NSMAL}(a,b) = \sum_{u=-S}^{S} \sum_{v=-S}^{S} \text{NSMAL}(a+u, b+v) \tag{12}$$

where $\text{NSMAL}_{\text{SAR}}$ and NSMAL_{MS} are used to represent the NSMAL of SAR and the MS image, respectively, and the fusion decision map can be defined as

$$\text{map}_D = (m \times \text{NSMAL}_{\text{MS}} \geq \text{NSMAL}_{\text{SAR}}) \tag{13}$$

where the weight coefficient $m > 1$ because the detailed information of the MS image is closer to the real situation. Therefore, the fused detail layer is represented by the following equation:

$$D_F = \text{map}_D \times D_{\text{MS}} + \left(1 - \text{map}_D\right) \times D_{\text{SAR}}. \tag{14}$$

III. Experiments

A. Experimental Design

1) Datasets: Twenty pairs of images are selected from the WHU-OPT-SAR dataset randomly, and the resolution of each pair of images is 1024 × 1024 pixels. Among them, the MS imagery is captured by the GF-1, including the red, green, blue (RGB), and near-infrared channel. The ground resolution is 2 m, and the resample is 5 m, and three bands of R, G, and B are mainly used in this article. The SAR imagery is captured by the GF-3 in the Fine Stripe II (FS II) mode, and the sampling resolution is 5 m [40].

2) Methods for Comparison: The BASHVS is compared with 16 methods. These methods are four conventional multiresolution decomposition frameworks: Curvelet [14], DTCWT [14], NSCT [14] and DWT [14]; a bivariate Gaussian convolution method over spatial-intensity domains: CBF [41]; a fusion method based on the optimization problem: gradient transfer and total variation minimization (GTF) [39]; four multiscale decomposition methods: HybridMSD [42], visual saliency map and weighted least square optimization (VSMWLS) [43]; multi-level Gaussian curvature filtering image decomposition (MLGCF) [44], and multiscale decomposition based on Least Squares optimization rolling guidance filtering (MSDLSRGF) [11]; and other six methods: convolutional sparsity based morphological component analysis (CSMCA) [45], CNN [46], Laplacian pyramid and sparse representation (LP-SR) [31], improved Laplacian energy and parameter adaptive dual-channel unit-linking pulse coupled neural network (WLEWSEML-NPCNN) [47], visual saliency features fusion (VSFF) [16], and dual-channel Rybak neural network (DRYNN) [48]. The parameters in the comparison methods are set according to the original texts.

3) Evaluation Indexes: Fused outputs undergo holistic evaluation via dual-mode methodology, integrating qualitative and quantitative assessment. Qualitative assessment appraises perceptual quality through localized detail fidelity and global structural integrity. Quantitative assessment measures image information content, spatial–spectral characteristics, and distortion levels via metric values. Ten indicators are selected in this article, and these indicators are demonstrated in Table I by their function categories. Entropy (EN) [14] evaluates the information amount of the fusion image; mutual information (MI) [49] is used to evaluate the gross information amount of the two source images included by the fusion image; average gradient (AG) [11] measures the fusion image's clarity; SF [11] measures an image's overall activity according to the change rate of the grayscale value; standard deviation (STD) evaluates the holistic contrast of the fusion image; spectral integrity degradation (SD) [11] quantifies spectral fidelity loss between fused and reference imagery; spectral angle mapper (SAM) [50] can quantify spectral fidelity of the fused image against the reference; Relative dimensionless global error in synthesis (ERGAS) [51] reflects the radiation differences between the fusion image and the reference image;

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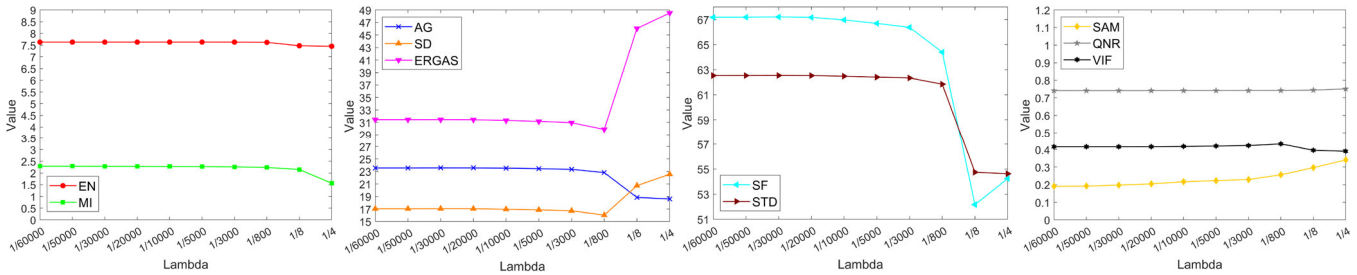


Fig. 4. Sensitivity analysis for the parameter λ. (a) Curve of EN and MI of lambda. (b) Curve of AG, SD, and ERGAS of lambda. (c) Curve of SF and STD of lambda. (d) Curve of SAM, QNR, and VIF of lambda.

TABLE I
QUANTITATIVE EVALUATION INDEXES AND THEIR EVALUATION DIMENSIONS

Index	Evaluation	Trend
Entropy	Information Quantity	The higher, the better(↑)
Mutual Information		The higher, the better(↑)
Average Gradient	Spatial Information	The higher, the better(↑)
Spatial Frequency		The higher, the better(↑)
Standard Deviation		The higher, the better(↑)
Spectral Integrity Degradation	Spectral Information	The lower, the better(↓)
Spectral Angle Mapper		The lower, the better(↓)
Relative dimensionless global error in synthesis		The lower, the better(↓)
Reference-Free Quality Evaluation	Degree of Distortion	The higher, the better(↑)
Visual Information Fidelity		The higher, the better(↑)

reference-free quality evaluation (QNR) [50] assesses integrated performance of fused visual data; visual information fidelity (VIF) [52] measures the effective visual information of the fusion image.

4) Parameter Setting: All image fusion methods are run on a computer equipped with 13th Gen Intel Core i7-13700 KF, using MATLAB R2023b. The following is the sensitivity analysis of the parameter λ in the SAD function. This analysis relies on EN, MI, AG, SF, STD, SD, SAM, ERGAS, QNR, and VIF for evaluation and displays the results in a curve chart. As known from Fig. 4, the values of EN, MI, AG, SF, STD, SD, SAM, ERGAS, QNR, and VIF all exhibit a decline in performance with the increase of λ. Comprehensively considering the model performance and the computational efficiency, λ is set to $1/30\,000$.

B. Results and Analysis

Select 20 pairs of images, fuse them through different fusion methods, and proceed with qualitative and quantitative evaluations of the fusion results.

1) Qualitative Evaluation: One of the image fusion results of the WHU-OPT-SAR dataset is shown in Fig. 5. The ground objects of this image mainly include buildings, roads, waters, vegetation, and bare land. As known from the results, the results of the four conventional multiresolution decomposition frameworks all have a low luminance with a little shadow on the roof, which means that they exhibit spectral distortion. The overall luminance of the fusion result of CBF is low, and the top of the building exhibits a phenomenon of spectral distortion. The GTF's result shows good spectral fidelity, but it has several speckle noises of the SAR image, which cause a building contour distortion in the upper left corner. The fusion results of Hybrid-MSD and CSMCA retain relatively complete spatial information, but still exist a phenomenon of slight spectral distortion. The fusion result of MSDLSRGF exhibits a phenomenon of losing the top information of the building. The fusion result of LP-SR has many white noises, and the contours of many ground objects are distorted. Although the VSFF can well retain the spatial information, the newly introduced structural information in the fused results fails to integrate effectively with the spectral information. Without compromising original geospatial features, the proposed fusion method effectively incorporates structural information from SAR images, thereby enriching spatial information while preserving spectral integrity. A red frame and a green frame are, respectively, used to select two areas in the image for further analysis. In the red frame, the pool contour in the fusion result of the BASHVS is the same as the original MS image, and the fusion result of the BASHVS has a higher contrast, which benefits visual interpretation. Waters in the other results are darker, exhibiting a phenomenon of spectral distortion, and the results are accompanied by white noise, having a poor visual effect. The luminance in the fusion results of the four conventional multiresolution decomposition frameworks is low, and the pool contour is distorted. Pool contours in the other comparison methods also exist in the phenomenon of spatial distortion. The pool contour of the LP-SR is accompanied by some white noise, causing spectral distortion, and the spatial information is distorted. The luminance in the fusion result of the VSFF is lower than the BASHVS. In the green frame, the BASHVS well retains the information of the road centerline and road edge line, enhancing the contrast. The results of other methods have a lower contrast, and the road is accompanied by noise, which also leads to the contour distortion of other ground objects. The road information in the results of GTF, MSDLSRGF, LP-SR, and Hybrid-MSD are lost, and the contours of the ground objects near the roads are distorted. The VSFF exhibits a coarser detail representation compared to the BASHVS.

2) Quantitative Evaluation: The relative performance of the fusion methods is evaluated by the information quantity, spatial information, spectral information, and distortion degree

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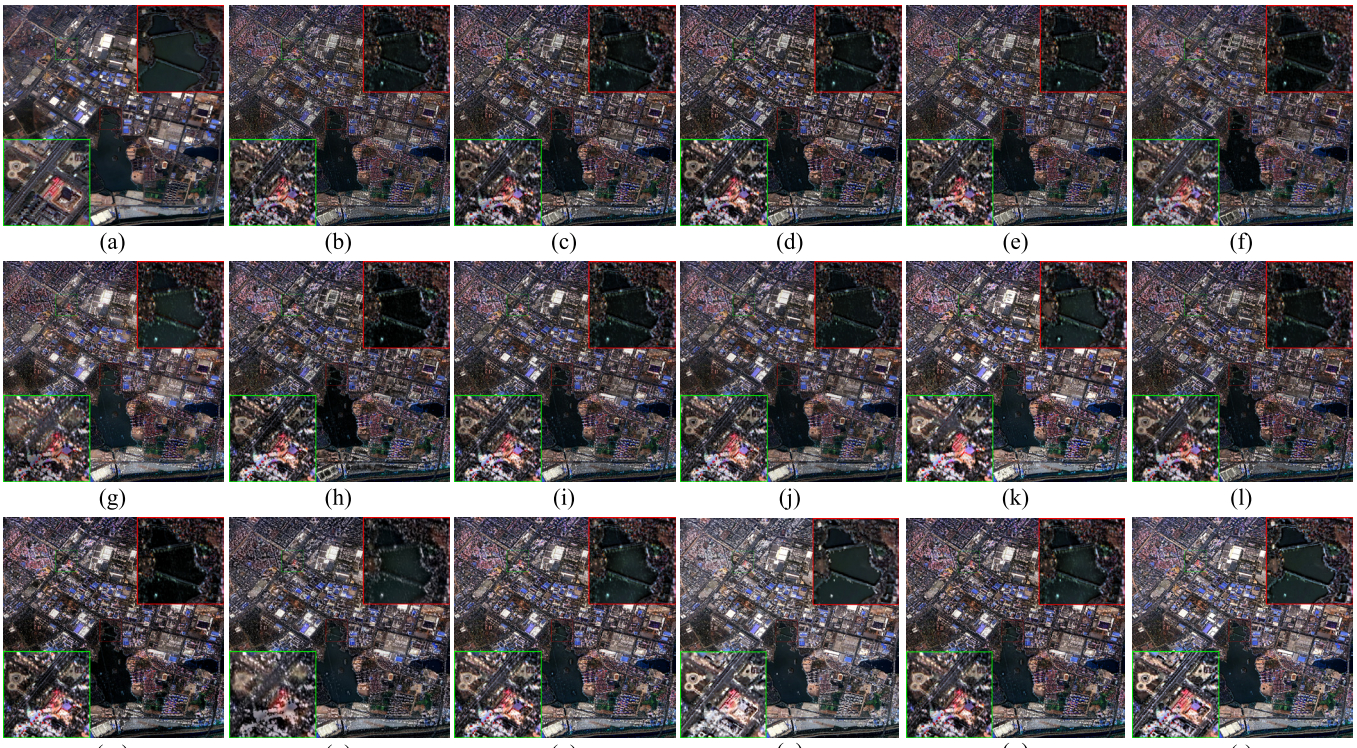


Fig. 5. Qualitative evaluation of 16 image fusion methods and the BASHVS on the WHU-OPT-SAR dataset. Elements are arranged left-to-right and top-to-bottom as: (a) MS image, (b) Curvelet, (c) DTCWT, (d) NSCT, (e) DWT, (f) CBF, (g) GTF, (h) Hybrid-MSD, (i) VSMWLS, (j) MLGCF, (k) MSDLSRGF, (l) CSMCA, (m) CNN, (n) LP-SR, (o) WLEWSEML-NPCNN, (p) VSFF, (q) DRYNN, and (r) BASHVS.

TABLE II

QUANTITATIVE METRICS FOR FUSED OUTPUTS OF VARIOUS METHODOLOGIES ON WHU-OPT-SAR

Methods	EN↑	MI↑	AG↑	SF↑	STD↑	SD↓	SAM↓	ERGAS↓	QNR↑	VIF↑
Curvelet	7.325	1.288	20.569	54.570	49.612	35.600	0.373	68.747	0.379	0.300
DTCWT	7.331	1.293	20.059	53.672	50.506	35.767	0.384	69.165	0.393	0.332
NSCT	7.346	1.424	20.561	54.271	52.050	35.727	0.405	68.313	0.373	0.359
DWT	7.327	1.221	21.134	56.994	51.001	36.359	0.388	70.656	0.407	0.296
CBF	7.307	1.516	18.596	48.378	46.731	35.697	0.399	75.585	0.403	0.314
GTF	7.489	1.228	18.521	49.273	52.024	32.263	0.279	56.572	0.345	0.362
Hybrid-MSD	7.382	1.414	<u>22.179</u>	60.735	54.200	38.884	0.622	79.223	0.360	0.296
VSMWLS	7.426	1.483	20.982	56.510	53.276	32.455	0.355	59.087	0.396	0.316
MLGCF	7.434	1.501	20.973	59.107	54.378	29.458	0.350	53.155	0.461	0.295
MSDLSRGF	7.606	1.668	20.890	56.787	58.578	25.237	<u>0.264</u>	47.487	0.531	0.325
CSMCA	7.347	1.512	19.461	51.676	49.819	36.841	0.291	70.882	0.355	**0.429**
CNN	7.218	1.339	20.401	55.594	56.634	38.115	0.566	76.358	0.412	0.344
LP-SR	7.550	1.185	20.068	51.743	56.448	34.997	6.484	58.367	0.315	0.399
WLEWSEML-NPCNN	7.596	1.456	21.839	57.773	58.110	25.566	0.330	43.325	0.458	0.333
VSFF	**7.682**	<u>2.121</u>	21.652	<u>61.306</u>	<u>63.708</u>	<u>21.400</u>	3.164	<u>37.127</u>	<u>0.628</u>	0.310
DRYNN	7.578	1.419	20.652	54.718	56.700	25.747	0.318	44.303	0.462	0.335
BASHVS	<u>7.628</u>	**2.162**	**23.894**	**67.689**	**63.922**	**18.125**	**0.212**	**32.010**	**0.747**	<u>0.425</u>

of the fusion result. The specific indicator values of each fusion method are listed in Table II. Each indicator value is the average value of the 20 pairs of images, and the indicator results are stable. The optimal value of the indicators is expressed in bold; the suboptimal value is underlined.

As known from Table II, the BASHVS has the best performance in the indicator evaluations of spatial information (AG, SF, and STD) and spectral information (SD, SAM, and ERGAS). The MI and QNR of the BASHVS are, respectively, 0.041 and 0.119 higher than the suboptimal method. The EN and VIF of the BASHVS are, respectively, only 0.054 and 0.004 lower than the optimal value. Therefore, the BASHVS demonstrates superior performance. For SD, SAM, and ERGAS, the values of BASHVS are, respectively, 3.275,

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TABLE III
PROCESSING TIME RECORD OF EACH FUSION METHOD

Method	Curvelet	DTCWT	NSCT	DWT	CBF	GTF	Hybrid-MSD	VSMWLS	MLGCF
Time/s	2.60	0.83	9.73	0.36	76.66	22.64	8.69	32.64	264.87
Method	**MSDLSRGF**	**CSMCA**	**CNN**	**LP-SR**	**WLEWSEML-NPCNN**	**VSFF**	**DRYNN**	**BASHVS**	
Time/s	117.09	1654.17	492.45	3172.80	1150.76	8.99	83.89	5.67	

0.052, and 5.117 lower than those of the suboptimal method, showing that BASHVS has higher spectrum fidelity. VSFF is a fusion method proposed recently, which is the best method except for the BASHVS, gaining suboptimal results on MI, SF, STD, SD, ERGAS, and QNR. On AG, SF, and STD, the VSFF is, respectively, 2.242, 6.383, and 0.214 lower than the BASHVS.

3) Time Complexity Analysis: The processing time of each fusion method is recorded in Table III. As known from Table III, the calculation time of Curvelet, DTCWT, and DWT is less than 3 s, making them less time-consuming. However, the mathematical models of these three methods are simple, and as known from Table II, the performances of these three methods are ordinary.

Although the mathematical models of MLGCF, MSDL-SRGF, CSMCA, CNN, LP-SR, and WLEWSEML-NPCNN are more complex, the time-consuming of these methods is all longer than 1 min, and their performances are ordinary. Among them, LP-SR is the longest time-consuming. The BASHVS needs an average time of 5.67 s to fuse a pair of images, which is less than the time-consuming of NSCT, CBF, GTF, Hybrid-MSD, VSMWLS, MLGCF, MSDLSRGF, CSMCA, CNN, LP-SR, WLEWSEML-NPCNN, VSFF, and DRYNN.

C. Ablation Study and Fusion Performance Analysis

1) Different Rules of Decomposition and Fusion: Three different fusion method variations are formed by modifying decomposition rules and fusion rules. The advantage of the BASHVS is verified by these variations. In variation 1, the decomposition rule is revised to rolling guide filtering (RGF); in variation 2, the fusion rule of the base layer is revised to the weighted sum method, i.e., $\mathrm{BF} = 0.5 \times B_{\mathrm{SAR}} + 0.5 \times B_{\mathrm{MS}}$; in variation 3, the decomposition rule is revised to RGF and the fusion rule of the base layer is revised to Average. Table IV lists the specific structures of each variation. Fig. 6 shows the fusion results of each variant method. Table V shows the average indicator values of 20 pairs of images fused by each variant method.

As known from Fig. 6, the fusion results of the three variant methods and the BASHVS all retain detailed information on the ground objects. However, the brightness of variations 2 and 3 is lower, and poor in visual effect. Variation 1 keeps the SHVS proposed in this article, so it has normal brightness. However, the building in the red frame is affected by noise. The building in the red frame of the BASHVS contains less noise, and the brightness is enhanced. As known from Table V, BASHVS demonstrates superior efficacy, securing the optimal indexes across spectral–spatial metrics.

TABLE IV
STRUCTURE OF THE BASHVS AND THREE VARIANTS

Methods	Decomposition	Base Layer
Variant1	RGF	SHVS
Variant2	SAD	Average
Variant3	RGF	Average
BASHVS	SAD	SHVS

The BASHVS excels in enhancing spatial information with minimal distortion and high fidelity.

2) Different Types of VS: The ablation experiment of VS is carried out to explain the necessity of applying PS and SS to the fusion work. The setting of the experiment is: 1) directly fuse the base layer of the MS and the SAR image without using VS; 2) only add PS into the fusion process; 3) only add SS into the fusion process; and 4) use PS and SS in the fusion process, i.e., the BASHVS. A 20 randomly selected pairs of images are used to perform the VS ablation experiment, and one set of the results is shown in Fig. 7. The average indicator value of the 20 pairs of images is recorded in Table VI.

As known from Fig. 7(a), when the VS is not taken into consideration, the fusion results have a lower brightness, the road information in the green frame is lost, and the building in the red frame contains noise. As known from Fig. 7(b), when only considering PS, the brightness of the image can be improved; however, the road information in the green frame is still lost, and the structure information of the building in the red frame is distorted. As known from Fig. 7(c), when only considering SS, the structural delineation of ground objects is significantly enhanced, albeit with attenuated luminance relative to Fig. 7(b). The BASHVS, which comprehensively considers PS and SS, not only enhances the brightness of fused images but also protrudes the contours of the ground objects in the fusion results. As known from Table VI, the BASHVS performs excellently in all the evaluation metrics, having the most stable performance. The EN value of the BASHVS is 0.108 higher than the suboptimal value; it has absolute dominance in spatial information evaluation indicators (AG, SF, and STD), which are 0.481, 0.835, and 0.207 higher than the suboptimal values, respectively. The BASHVS has

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TABLE V
INDEX VALUES OF THE BASHVS AND ITS THREE VARIANTS

	EN↑	MI↑	AG↑	SF↑	STD↑	SD↓	SAM↓	ERGAS↓	QNR↑	VIF↑
Variant1	7.603	**2.359**	20.220	55.092	59.616	**14.251**	0.258	**28.785**	0.738	0.394
Variant2	7.126	1.949	13.485	36.310	39.673	28.498	0.255	55.879	0.479	0.304
Variant3	7.289	1.440	17.849	49.832	44.827	26.955	0.298	52.988	0.598	0.311
BASHVS	**7.628**	2.162	**23.894**	**67.689**	**63.922**	18.125	**0.212**	32.010	**0.747**	**0.425**

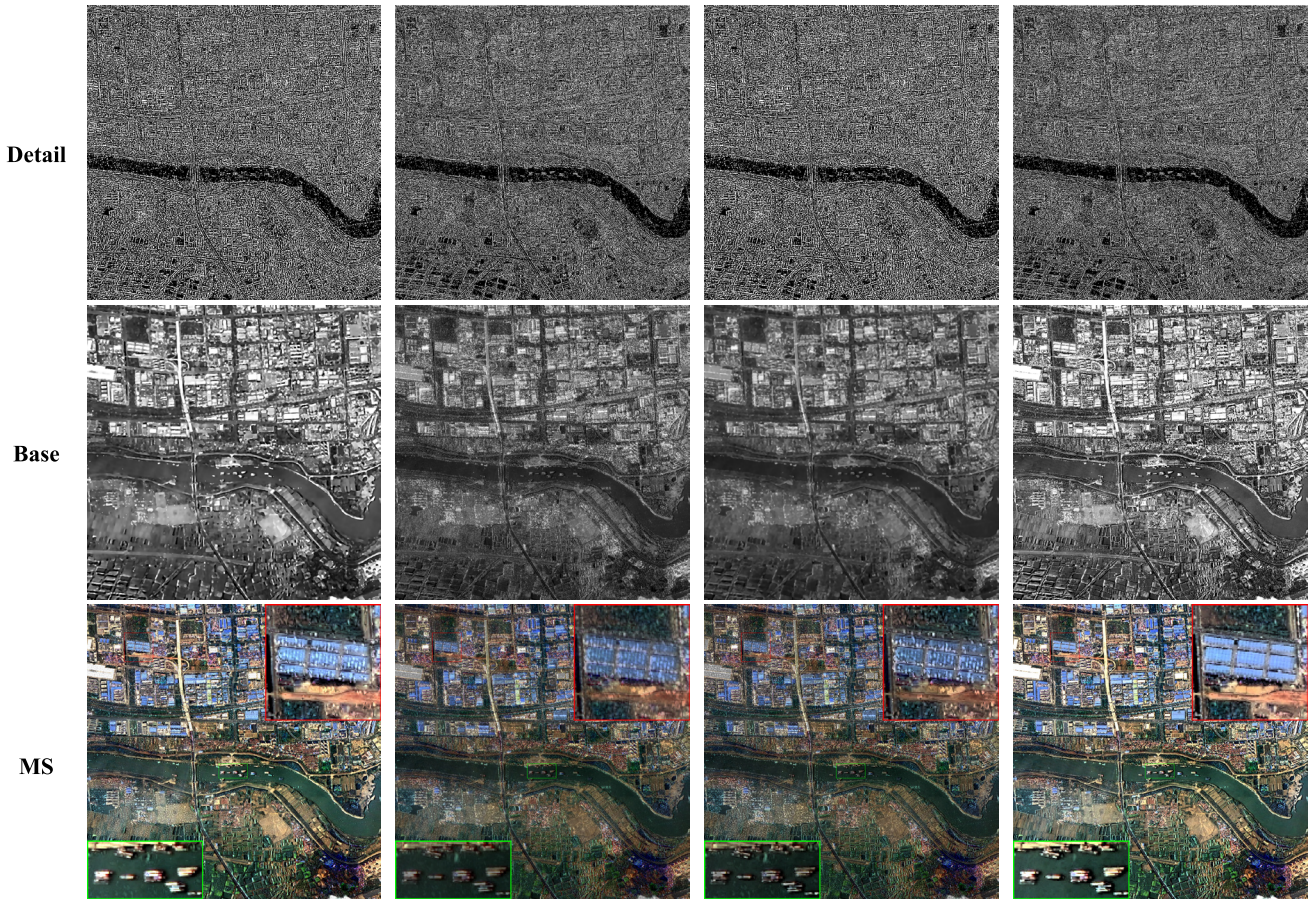


Fig. 6. Fusion result of the BASHVS and its three variants. The first variant is (a) RGF-SHVS, the second variant is (b) SAD-Average, the third variant is (c) RGF-Average, and the last one is (d) BASHVS. Row 1: detail layer fusion output, Row 2: base layer fusion output, Row 3: fused MS image.

TABLE VI
INDEX VALUES OF DIFFERENT TYPES OF VS

	EN↑	MI↑	AG↑	SF↑	STD↑	SD↓	SAM↓	ERGAS↓	QNR↑	VIF↑
No VS	7.500	1.250	18.169	48.327	51.521	30.077	0.211	53.168	0.377	0.335
PS	7.520	1.395	18.569	49.446	51.893	31.671	**0.174**	52.996	0.347	0.350
SS	7.491	**2.191**	23.413	66.854	63.715	**15.770**	0.258	**28.538**	**0.781**	**0.449**
BASHVS	**7.628**	2.162	**23.894**	**67.689**	**63.922**	18.125	0.212	32.010	0.747	0.425

great performance on MI, SD, SAM, ERGAS, QNR, and VIF. When only considering SS, the fusion results have a lower distortion degree and a better performance on SD and ERGAS. However, it has poor performance on EN and SAM, which explains that the performance is unstable when only SS is considered.

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TABLE VII
OA AND KAPPA OF LC CLASSIFICATION IN THE FIRST REGION

Method	MS	Curvelet	DTCWT	NSCT	DWT	CBF	GTF	Hybrid-MSD	VSMWLS
OA	98.003%	98.647%	97.809%	98.325%	98.131%	97.616%	98.067%	97.938%	97.809%
KC	0.973	0.982	0.970	0.977	0.975	0.968	0.974	0.972	0.970
Method	**MLGCF**	**MSDLSRGF**	**CSMCA**	**CNN**	**LP-SR**	**WLEWSEML-NPCNN**	**VSFF**	**DRYNN**	**BASHVS**
OA	99.162%	97.809%	98.389%	98.389%	98.067%	98.196%	**99.356%**	98.840%	99.227%
KC	0.989	0.970	0.978	0.978	0.974	0.976	**0.991**	0.984	0.990

TABLE VIII
OA AND KAPPA OF LC CLASSIFICATION IN THE SECOND REGION

Method	MS	Curvelet	DTCWT	NSCT	DWT	CBF	GTF	Hybrid-MSD	VSMWLS
OA	98.123%	96.768%	97.497%	97.393%	97.132%	94.421%	96.663%	96.455%	97.654%
KC	0.976	0.958	0.968	0.966	0.963	0.928	0.957	0.954	0.970
Method	**MLGCF**	**MSDLSRGF**	**CSMCA**	**CNN**	**LP-SR**	**WLEWSEML-NPCNN**	**VSFF**	**DRYNN**	**BASHVS**
OA	98.123%	96.820%	96.820%	93.796%	93.535%	96.663%	96.768%	97.393%	**98.488%**
KC	0.976	0.959	0.959	0.920	0.917	0.957	0.958	0.966	**0.980**

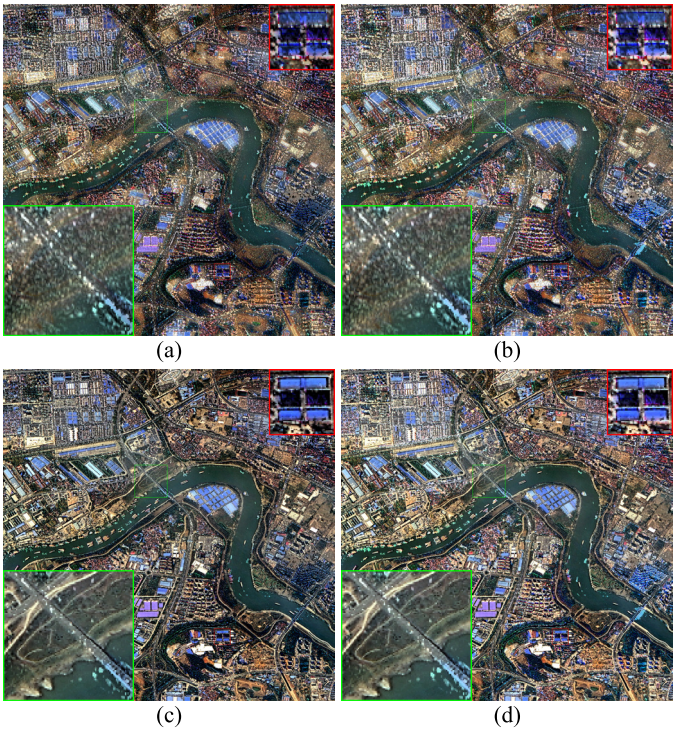


Fig. 7. Results of the VS ablation experiment. (a) No VS. (b) PS. (c) SS. (d) BASHVS.

D. LC Classification

To verify the practicality of the BASHVS, in this article, the random forest is used to carry out LC classification experiments on the images of three regions. The overall accuracy (OA) and the Kappa coefficient (Kappa) are the classification evaluation indicators. According to the high-resolution feature of the experimental data, the LC classification samples, used for training and verifying, are all randomly selected from the original MS images. Each LC type selects 50 training samples and 50 verifying samples separately. Identical sample sets and experimental conditions are maintained across LC classifications to preserve methodological rigor. The classification results of regions 1, 2, and 3 are separately shown in Figs. 8–10, and the cartographic correctness evaluation results are, respectively, shown in Tables VII–IX.

As known from Fig. 8, owing to spectral aliasing between built structures and transportation networks, cross-category misclassification persists across fusion methodologies. Furthermore, some buildings in the original MS image are misclassified as vegetation. MSDLSRGF misclassifies much vegetation as roads. In comparison, BASHVS has a better classification effect. However, VSFF has a more robust ability to distinguish between roads and buildings. As known from Fig. 9, BASHVS has the closest result to the real distribution of ground objects, protruding more vegetation. As a result of CNN, some bare land and vegetation are misclassified as water. In the results of LP-SR, many areas are misclassified as roads. As a result of Hybrid-MSD, the vegetation below is misclassified as roads. As known from Fig. 10, the classification result of BASHVS has a high degree of aggregation and clear boundaries, showing the best visual effect. Much vegetation in the original MS image is misclassified as buildings. Other methods exhibit fragmentation and exist in a phenomenon of misclassification. BASHVS has the best LC classification effect.

As known from Table VII, the OA of all the methods is higher than 95%, and the Kappa is all higher than 0.95. The OA and Kappa of VSFF are the highest, which are, respectively, 99.356% and 0.991, and increased by 1.353% and 0.018 against the source MS imagery. The classification result of BASHVS shows suboptimal performance, and the OA and Kappa are, respectively, 0.129% and 0.001 lower than VSFF. MLGCF has an OA value of 99.162% and its Kappa is 0.001 lower than BASHVS. The DTCWT, CBF,

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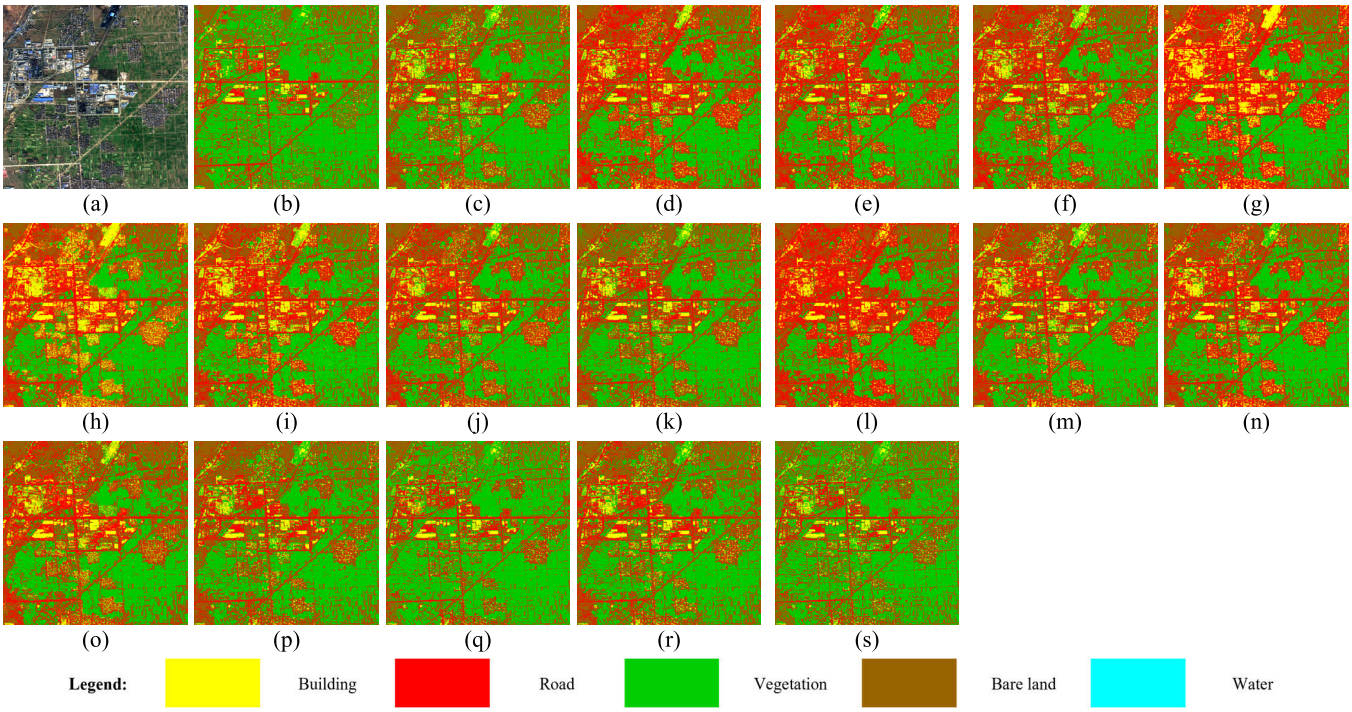


Fig. 8. LC classification results of different methods in the first region. (a) RGB image and elements are arranged left-to-right and top-to-bottom as: (b) MS image, (c) Curvelet, (d) DTCWT, (e) NSCT, (f) DWT, (g) CBF, (h) GTF, (i) Hybrid-MSD, (j) VSMWLS, (k) MLGCF, (l) MSDLSRGF, (m) CSMCA, (n) CNN, (o) LP-SR, (p) WLEWSEML-NPCNN, (q) VSFF, (r) DRYNN, and (s) BASHVS.

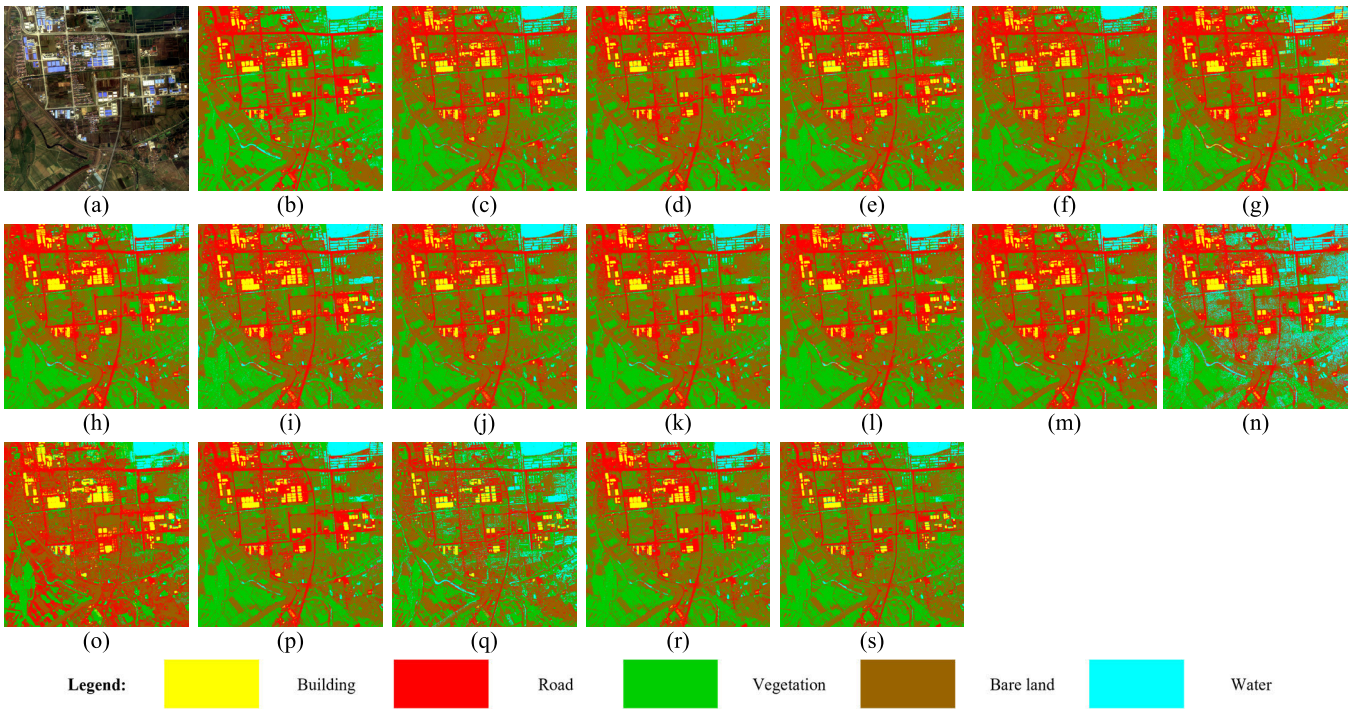


Fig. 9. LC classification results of different methods in the second region. (a) RGB image and elements are arranged left-to-right and top-to-bottom as: (b) MS image, (c) Curvelet, (d) DTCWT, (e) NSCT, (f) DWT, (g) CBF, (h) GTF, (i) Hybrid-MSD, (j) VSMWLS, (k) MLGCF, (l) MSDLSRGF, (m) CSMCA, (n) CNN, (o) LP-SR, (p) WLEWSEML-NPCNN, (q) VSFF, (r) DRYNN, and (s) BASHVS.

Hybrid-MSD, VSMWLS, and MSDLSRGF all exhibit spectral attenuation relative to the source MS imagery, indicating fidelity degradation in fused outputs across evaluated methods. As known from Table VIII, BASHVS has the best OA and Kappa, which are increased by 0.365% and 0.004 against the source MS imagery. MLGCF has the same OA as the original MS image. Hence, both outputs achieve suboptimal cartographic correctness. The cartographic correctness of the other methods is all lower than the cartographic correctness of the original MS image. As known from Table IX, the cartographic correctness of BASHVS is optimal, and VSFF is suboptimal. The OA and the Kappa of the BASHVS are, respectively, increased by 1.562% and 0.022 against the source MS imagery.

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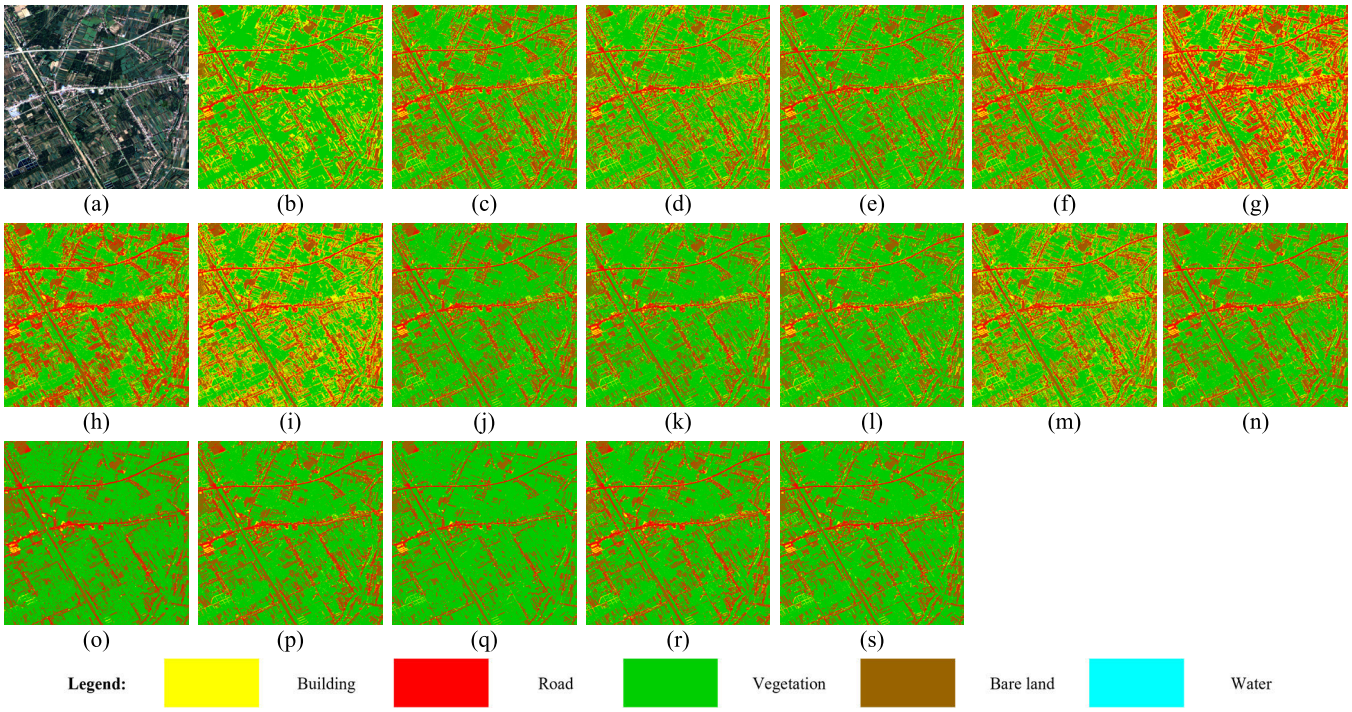


Fig. 10. LC classification results of different methods in the third region. (a) RGB image and elements are arranged left-to-right and top-to-bottom as: (b) MS image, (c) Curvelet, (d) DTCWT, (e) NSCT, (f) DWT, (g) CBF, (h) GTF, (i) Hybrid-MSD, (j) VSMWLS, (k) MLGCF, (l) MSDLSRGF, (m) CSMCA, (n) CNN, (o) LP-SR, (p) WLEWSEML-NPCNN, (q) VSFF, (r) DRYNN, and (s) BASHVS.

TABLE IX
OA AND KAPPA OF SS CLASSIFICATION IN THE THIRD REGION

Method	MS	Curvelet	DTCWT	NSCT	DWT	CBF	GTF	Hybrid-MSD	VSMWLS
OA	97.100%	97.267%	97.044%	97.267%	97.379%	96.096%	96.542%	96.263%	97.211%
KC	0.959	0.961	0.958	0.961	0.963	0.944	0.950	0.947	0.960
Method	**MLGCF**	**MSDLSRGF**	**CSMCA**	**CNN**	**LP-SR**	**WLEWSEML-NPCNN**	**VSFF**	**DRYNN**	**BASHVS**
OA	96.988%	97.267%	96.932%	97.267%	95.538%	97.658%	98.160%	97.434%	**98.662%**
KC	0.957	0.961	0.956	0.961	0.936	0.967	0.974	0.963	**0.981**

IV. CONCLUSION

Remote sensing image fusion fundamentally synthesizes complementary components from multimodal sources to enhance geospatial semantics extraction. For the question that current fusion methods do not consider the model difference between MS and SAR images, which leads to information inconsistency and serious interference by speckle noise, a fusion algorithm (BASHVS), based on the bidirectional aggregation of the saliency in the HVS, is proposed. This algorithm factorizes imagery into structural bases and detail manifolds while getting the diffusion function values in eight directions. It is need to add these diffusion function values into the NSMAL algorithm when fusing the detail layer. For the base layer, first, an MSD is required, and then the result of decomposition is transformed into the state of the HVS to separately calculate its PS from small scales to large scales and its SS from large scales to small scales, fusing the base layer based on the PS and the SS. Compared with 16 fusion methods, the BASHVS performs the best on eight evaluation indicators. The fusion result of the BASHVS has a preferable contrast ratio, retains relatively complete information about the ground objects, and does not introduce excess noise. By counting the time complexity of each method, it is found that the BASHVS is less time-consuming, so it can be put into a real-time application. Ablation experiments prove that the decomposition rules and fusion rules of the BASHVS contribute to improved fusion performance, and the proposed SHVS module can improve visual effects. In the LC classification experiments of three regions, BASHVS achieves significant superiority. On the whole, the BASHVS not only enhances the space information, retains spectral information, and reduces the degree of distortion, but also further elevates the precision of the LC classification.

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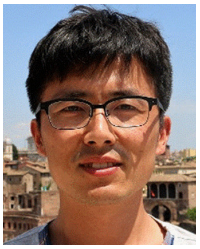
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